

**Near-fault Ground Motion Prediction, Site Effects and
Wave Propagation at Regional Scale for a Shallow
Earthquake in Southeastern France**
**Évaluation du mouvement sismique en champ proche, prise
en compte des effets de site et de propagation régionale
pour un séisme superficiel dans le Sud-est de la France**

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0.1 Overview of Chapter One

In this chapter, I employ a physics-based dynamic rupture method to model earthquake rupture while simultaneously simulating surface ground motion.

As explained in the previous chapter, the primary challenge in simulating near-fault ground motion (NFGM) for the 2019 M_w 4.9 Le Teil earthquake is the lack of near-field recorded data. This challenge is not unique to this specific event; generally, for small-to-moderate earthquakes, strong-motion records are often sparse. Because the seismic magnitude is relatively low, the waves attenuate before reaching distant strong-motion seismometers with sufficient amplitude. This limitation is also why the kinematic model by [10] primarily relied on regional stations operating in lower frequency ranges.

Despite this lack of empirical data, accurate estimations of expected ground motion are crucial for seismic hazard management and risk assessment. These estimations directly inform critical decision-making processes, such as enforcing local building codes or designing sensitive infrastructure like nuclear facilities. In an ideal world with unlimited scientific funding, seismologists could deploy dense networks of strong-motion seismometers worldwide with perfect spatial resolution. The problem would be solved: we could use the precise records to generate comprehensive seismic hazard maps, and policymakers could hopefully make wise decisions. However, we do not live in an ideal world. Even if unlimited funding existed, I doubt such a network could be easily realized; at least in Europe, strict regulations regarding personal privacy violations concerning the installation of seismometers would likely get in the way first.

It is within this context that I frame my research on near-fault ground motion simulation. By utilizing computers, I am building a “digital twin” of the earthquake rupture, wave propagation, and surface ground motion. In this digital twin, I am omnipotent like ‘God’—I can place virtual seismogram stations wherever I want and as many as possible. Although the computational costs increase as the number of virtual stations grows, it is peanuts compared to the financial and logistical burden of operating real physical stations.

For sure, in the digital world that I created, I must make necessary abstractions of physics that known to humankind. I need to choose which physical complexities to include and which to omit, either because the required data is unavailable or because they do not significantly contribute to the specific problem I am investigating. For instance, I must exclude topography; although it significantly impacts near-fault ground motion, I lack the high-resolution 3D velocity structure in the study area required to accurately model topographic effects.

To borrow Stephen Hawking’s definition, a good model is one that produces measurable predictions, for example Sir Isaac Newton’s model of gravity is a fantastic model, which successfully predicts the observable orbits of celestial bodies. In this regard, because there are no directly usable near-fault seismic records for the Le Teil earthquake, I must validate my predictions of near-fault ground motion (NFGM) indirectly by comparing them with

those derived from empirical models, like Ground Motion Prediction Equations (GMPEs) or Ground Motion Models (GMMs).

Regarding the challenge of predicting near-fault ground motion, it is important to address this industry-standard method. GMPEs are empirical models calibrated using historical seismic data. However, because the training data for low-to-moderate magnitude, extremely shallow earthquakes—such as the Le Teil event—is notoriously sparse [12], it is entirely reasonable to question the validity and accuracy of the NFGM predictions derived from standard GMPEs for this specific scenario.

In this chapter, I explain my dynamic model in detail, utilizing a 3D elastic medium and a planar fault geometry. I outline the physical theories applied, including the linear slip-weakening friction law and the Mohr-Coulomb failure criterion, assuming an Andersonian state of stress, which means for a reverse-faulting mechanism, the minimum principal stress is vertical.

I also detail the methodology for calculating the initial stress and frictional parameters (i.e., dynamic parameters) from a data-inverted kinematic model. First, I replace the original triangular-shaped source time function of the kinematic model with a dynamically compatible, regularized Yoffe-type source time function. Following this, I derive the dynamic parameters. In Section 1.2.2, I present the mathematical derivation for estimating these dynamic parameters without solving the elastodynamic equations. For this analytical approach, I must assume three parameters: the prestress ratio at the optimally oriented angle (R_{opt} , which describes the initial stress level and its proximity to failure), the static friction coefficient (μ_s), and the cohesion (c). Additionally, I present the method for calculating dynamic parameters by explicitly solving the elastodynamic equations; this is the primary method utilized in the attached manuscript (Section ??).

Finally, in the paper manuscript, I introduce physical complexity to the earthquake source parameters by enforcing a strain-consistency condition [3] and applying a fractal template [9]. I then compare the near-fault ground motion (NFGM) generated by these dynamic rupture models against three empirical Ground Motion Prediction Equations (GMPEs). These comparisons are evaluated using Spectral Acceleration (SA) at 0.5 Hz and 1 Hz, a critical metric in earthquake engineering that characterizes the intensity and potential impact of ground shaking on building structures.

Chapter 1

Physics-based Dynamic Ruptrure Modelling

1.1 Latest Kinematic Model of the 2019 Le Teil Earthquake

I base my dynamic modeling on the most recent kinematic model, which was derived from a joint seismic and geodetic inversion (for full inversion details, refer to Godano et al. [10]).

This kinematic model is defined on a planar fault parameterized by 60 subfaults. Each subfault measures 0.5 km along both strike and dip, covering an area of 0.25 km². The computational grid comprises 12 subfaults along the strike and 5 along the dip, resulting in a total fault length of 6 km and a down-dip width of 2.5 km. The fault strikes 47° relative to the north and dips 57° to the southeast.

Figure 1.1 illustrates the fault geometry. The number in the center of each subfault indicates its specific moment magnitude. The total seismic moment of the entire fault is calculated as the sum of the seismic moments from all individual subfaults, yielding an overall event magnitude of M_w 4.95.

Figure 1.2 displays the interpolated kinematic model generated with different levels of spatial smoothing. The inverted triangles denote the centers of the individual subfaults, while the arrows indicate the direction of slip. The magnitude of the slip is represented by the color bar.

For the conversion to Regularized Yoffe Source Time Functions (Section ??) and all subsequent dynamic modeling, I exclusively use the strongly smoothed version of the kinematic slip model. This choice is made because the stress drop derived from the strongly smoothed kinematic model is smaller and physically more realistic.

1.1.1 Conversion to Regularized Yoffe Source Time Functions

The original slip rate function of the kinematic model consists of three overlapping isosceles triangular functions. To make the slip rate function physically consistent, I convert these into regularized Yoffe slip rate functions using the formulations by Tinti et al. [17].

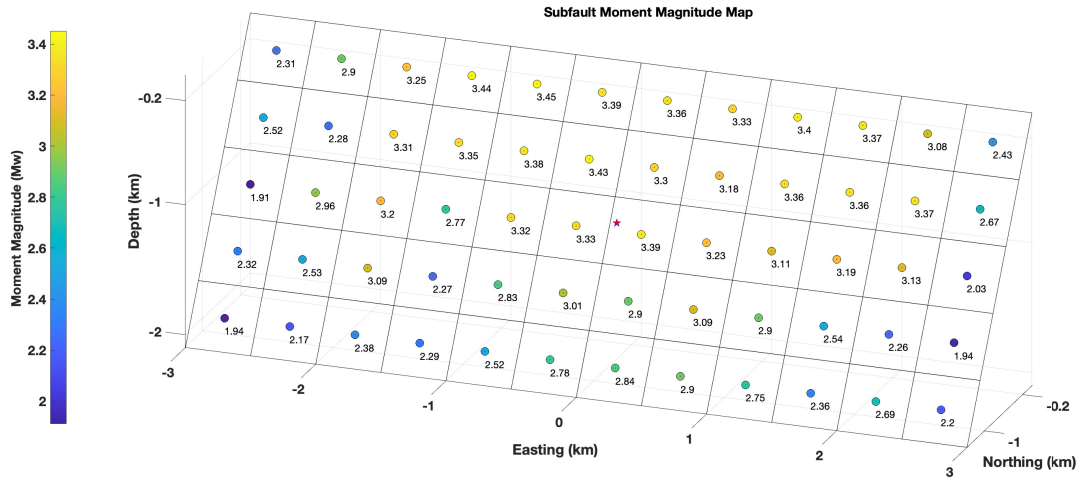
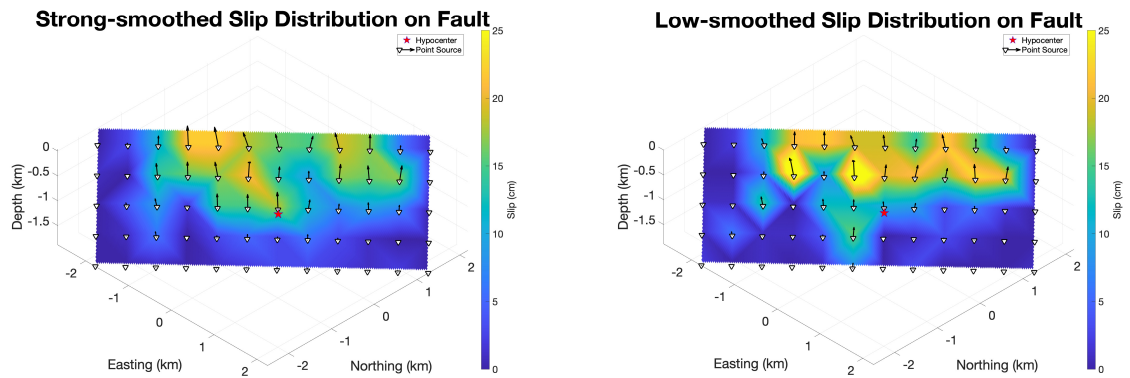


Fig. 1.1 Configuration of the kinematic model. The red star indicates the hypocenter used in the dynamic rupture simulations. The fault length is 6 km, the fault width is 2.5 km, and the dip angle is 57° . The color bar indicates the moment magnitude of each subfault.



(a) Strongly smoothed kinematic model.

(b) Weakly smoothed kinematic model.

Fig. 1.2 Comparison of the continuous kinematic slip models generated using (a) strong and (b) weak spatial smoothing. The slip magnitude is indicated by the color bar, and the slip direction is denoted by the arrows. The inverted triangles represent the centers of the subfaults, and the star indicates the hypocenter.

The source time function (STF) used in kinematic modeling, also known as the slip rate function, describes how fault slip evolves over time. I show an example of a constructed slip rate function for a single subfault alongside its corresponding regularized Yoffe slip rate function in Figure 1.3. To construct the regularized Yoffe function, only three parameters are required: T_{acc} , τ_R , and D_{max} . T_{acc} is the positive acceleration time, which defines when the slip rate reaches its maximum (V_{peak}). τ_R is the total rupture duration, and D_{max} is the final slip. By definition, all overlapping isosceles triangular STFs in this setup have a total duration of 0.6 seconds.

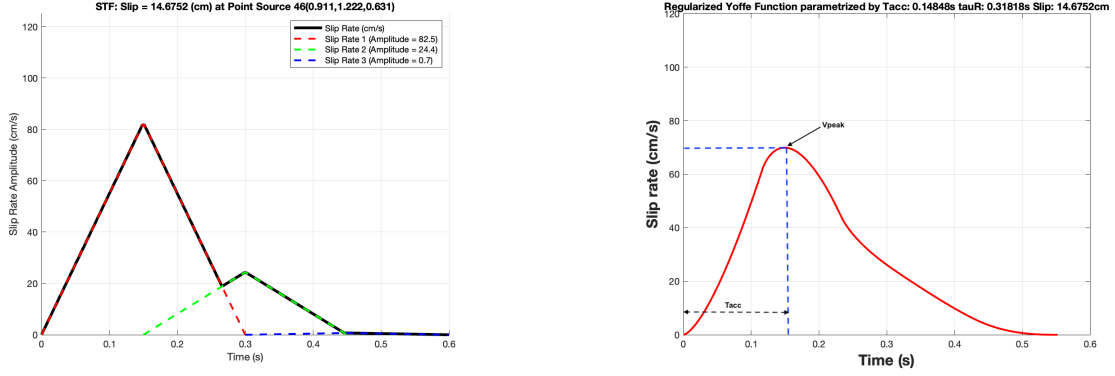


Fig. 1.3 Example of a constructed slip rate function for the same subfault, utilizing (a) three overlapping isosceles triangles and (b) a regularized Yoffe slip rate function.

Figure 1.4 shows the slip evolution curves, where slip is computed by integrating the area under the slip rate functions. The total duration for the overlapping isosceles triangle slip rate functions is 0.6 s. In contrast, the duration for the regularized Yoffe slip rate functions is $2 \cdot \tau_s + \tau_R$ due to convolution, where τ_s is T_{acc} divided by 1.27 [17].

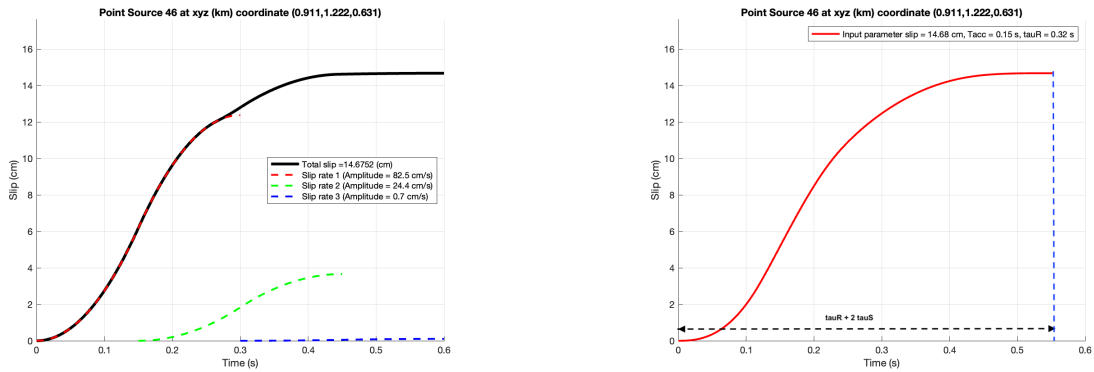


Fig. 1.4 Constructed slip evolution curve for the same subfault from overlapped triangular source time functions (left), and from regularized Yoffe source time functions (right).

Figure 1.5 compares the slip evolution derived from the overlapping triangular STFs and the converted regularized Yoffe STFs across all 60 subfaults. I observe some discrepancies between the two models, which actually represents a physical improvement in the modeling. The overlapping triangular STFs occasionally produce rapid, stepwise increases in slip,

implying that the rupture briefly stalls at a subfault before unphysically reactivating. This is an artificial side effect inherent to the triangular source time function. In contrast, the slip evolution produced by the regularized Yoffe function is gradual and continuous, accurately reflecting a realistic rupture process characterized by initiation, acceleration, and deceleration phases.

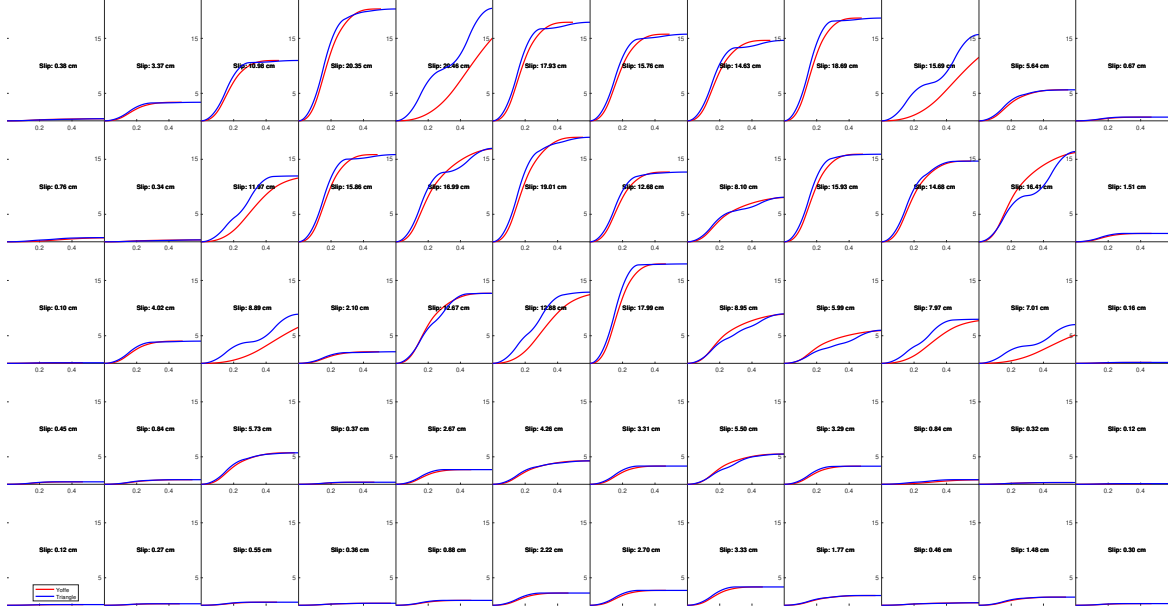
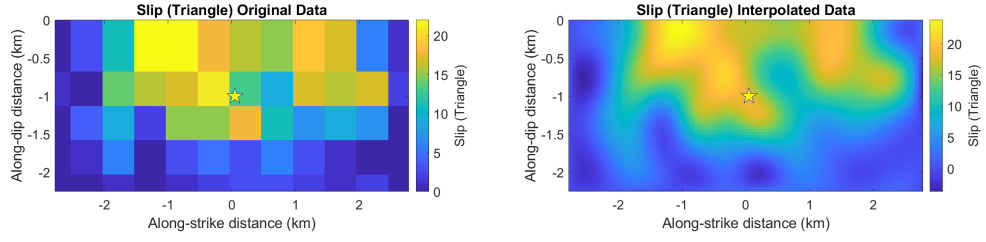


Fig. 1.5 Comparison of slip evolution derived from the overlapping triangular STF and the converted regularized Yoffe STF.

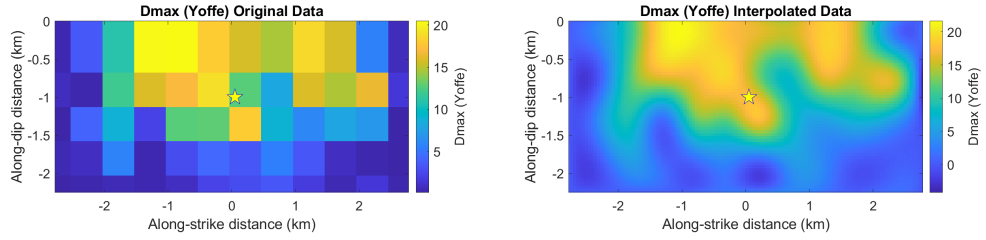
Figure 1.6a displays the original and linearly interpolated slip distributions using the triangular source time function, while Figure 1.6b shows the original and linearly interpolated slip distributions using the regularized Yoffe source time function. The differences between the two are negligible. In the next section, I explain how I use this D_{max} —the linearly interpolated slip built with the regularized Yoffe source time function—to estimate D_c (Section 1.2.5) and introduce heterogeneities into my model.

1.2 Estimating Dynamic Parameters without Solving Elastodynamic Equations

Determining the initial stress conditions is essential for performing dynamic rupture simulations. In my model, the only source of heterogeneity is D_{max} , which is derived from the converted kinematic slip. Here, I present a theoretical method to calculate the initial stress state. This approach assumes three known parameters: the prestress ratio at the optimally oriented angle (R_{opt}), the static friction coefficient (μ_s), and the cohesion (c). The R_{opt} parameter can be correlated with D_{max} . For example, I assume that they are positively correlated, meaning that areas with greater slip indicate a stress state that is more favorable to dynamic rupture. The value of μ_s is typically chosen based on the local rock unit, while cohesion



(a) Final slip in the kinematic model across the fault plane before and after interpolation.



(b) D_{max} across the fault plane before and after interpolation.

Fig. 1.6 Comparison of slip distributions across the fault plane before and after interpolation. The yellow star denotes the hypocenter

is generally treated as a tuning parameter, determined through trial and error to effectively control the dynamic rupture process.

1.2.1 Model Setup

Figure 1.7 displays the Mohr-Coulomb diagram detailing the dynamic parameter setup for the Le Teil earthquake. Under these stress conditions, the optimally oriented angle (ϕ^{opt}) between the maximum principal stress (S_1) and the slip direction is less than 45° . This is because the static friction coefficient for limestone is likely less than 0.6, which corresponds to a specific ϕ^{opt} around 45° based on Equation 1.2.1. Additionally, the fault dip angle is 57° .

The initial stress is described by a rank-2 stress tensor at each mesh grid point. Rank-2 tensors are typically represented as matrices. The corresponding stress matrix is symmetric and is parameterized by six components:

$$\begin{bmatrix} b_{xx} & b_{xy} & b_{xz} \\ & b_{yy} & b_{yz} \\ & & b_{zz} \end{bmatrix}$$

By rotating the stress tensor, I can find the three principal stresses ($S_1 > S_2 > S_3$), where the shear stress is exactly zero:

$$\begin{bmatrix} S_1 & & \\ & S_2 & \\ & & S_3 \end{bmatrix}$$

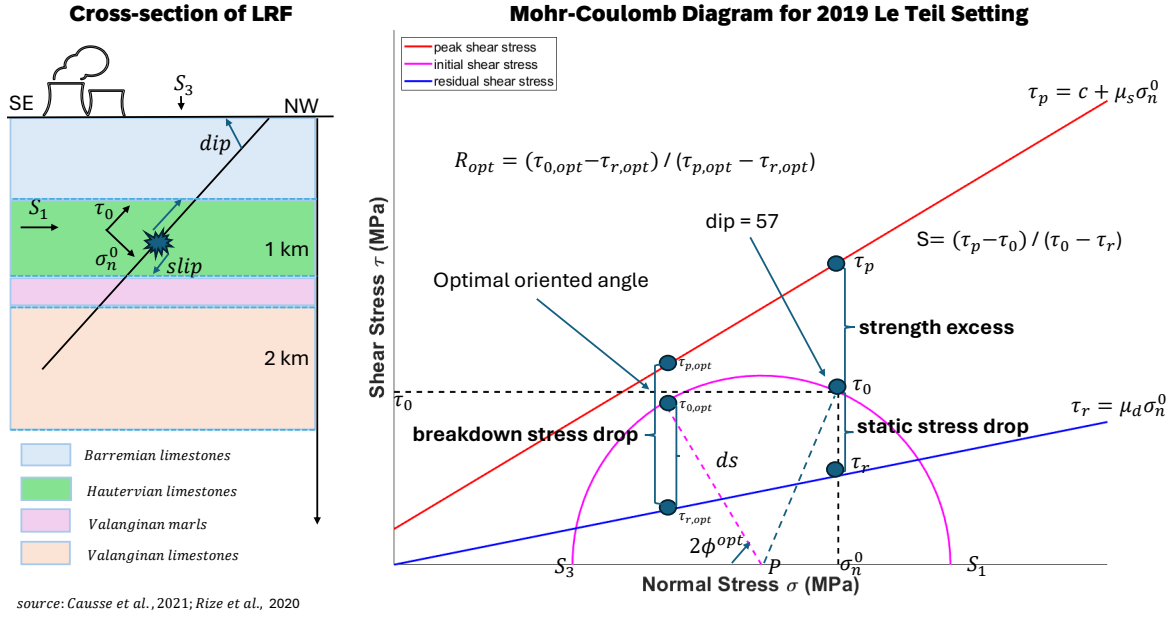


Fig. 1.7 Mohr-Coulomb diagram for the 2019 Le Teil, France earthquake setting.

Conversely, I can compute the six stress tensor components from the principal stresses. I prefer working with principal stresses because they relate directly to the Mohr-Coulomb circle.

Anderson's theory of faulting assumes that one of the principal stresses is vertical. For reverse faulting focal mechanisms, S_3 is vertical, can be approximated as $\rho g z$,

and S_1 is the maximum horizontal stress (perpendicular to S_3), as shown in Figure 1.7. Below, I demonstrate how the principal stresses are calculated theoretically based on the Mohr-Coulomb failure criterion.

In Figure 1.7, ϕ^{opt} is the optimal angle, μ_s is the static friction coefficient, P is the center (locus) of the optimally oriented Mohr's circle, and ds is its radius. The difference between the maximum and minimum principal stresses is twice ds :

$$\phi^{opt} = \frac{\pi}{4} - \frac{1}{2} \arctan(\mu_s)$$

$$ds = \frac{1}{2}(S_1 - S_3)$$

$$P = \frac{1}{2}(S_1 + S_3)$$

The parameter $\nu = 0.5$ [19] defines the intermediate principal stress relative to the maximum and minimum stresses, expressed as $\nu = \frac{S_2 - S_3}{S_1 - S_3}$. Therefore:

$$\begin{cases} S_1 = P + ds \\ S_2 = P - ds + 2 \cdot \mathbf{v} \cdot ds \\ S_3 = P - ds \end{cases}$$

I define the initial stress state for the optimal angle ϕ^{opt} as:

$$\tau_{0,opt} = ds \cdot \sin(2 \cdot \phi^{opt})$$

$$\sigma_{n,opt}^0 = P - ds \cdot \cos(2 \cdot \phi^{opt})$$

In Figure 1.7, the red line denotes peak stress and the blue line denotes residual stress. With c as cohesion and μ_d as the dynamic friction coefficient, I have:

$$\tau_{p,opt} = c + \mu_s \cdot \sigma_{n,opt}^0$$

$$\tau_{r,opt} = \mu_d \cdot \sigma_{n,opt}^0$$

The prestress ratio R_{opt} is the ratio of the static stress drop ($\tau_0 - \tau_r$) to the breakdown stress drop ($\tau_p - \tau_r$), analogous to the T parameter in Aochi & Ulrich, 2015 [4]:

$$R_{opt} = \frac{1}{1 + S_{opt}} = \frac{\tau_{0,opt} - \tau_{r,opt}}{\tau_{p,opt} - \tau_{r,opt}}$$

The S parameter is the ratio between stress excess (peak stress - initial stress) and dynamic stress drop (initial stress - residual stress). Before performing dynamic rupture simulations, I can predict rupture growth based on the model's S profiles.

S is defined as ([1][7]),

$$S = \frac{\tau_p - \tau_0}{\tau_0 - \tau_r} \quad (1.1)$$

$$\begin{cases} S < 0 & \text{unfavorable} \\ 0 \leq S \leq 1 & \text{favorable} \\ S >> 1 & \text{unfavorable} \end{cases} \quad (1.2)$$

When the S -ratio is fixed, a higher μ_s results in greater absolute initial normal and shear stresses (a larger Mohr's circle). Figure 1.8a illustrates how the optimally oriented angle ϕ varies with changes in μ_s . I observe that ϕ is highly sensitive to μ_s ; a small variation in ϕ corresponds to a substantial shift in the necessary μ_s . This sensitivity introduces a challenge because the dip angle derived from kinematic inversions is often poorly constrained [8]. For example, Aochi and Tsuda [3] utilized a dip angle of 50° , which generated a much higher stress level and a more favorable S -parameter compared to the 57° dip used in my setup.

Figure 1.8b visually reinforces that a higher μ_s expands the Mohr's circle, thereby dictating higher absolute stresses.

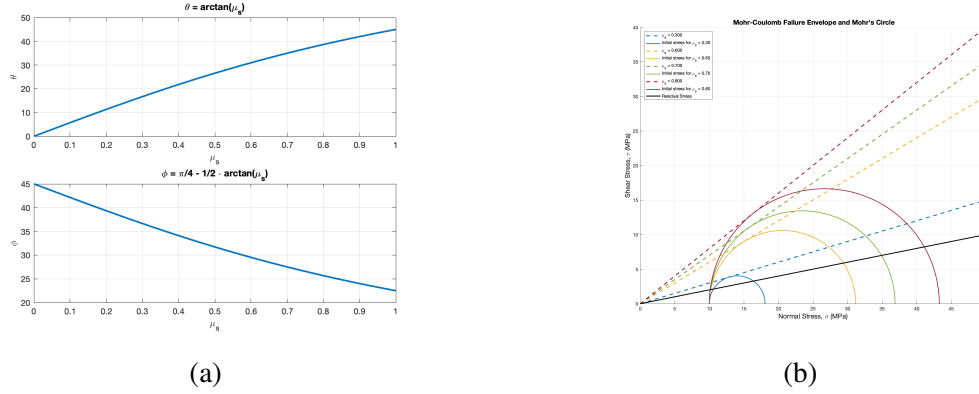


Fig. 1.8 (a) ϕ and θ as a function of μ_s . (b) Mohr's circle expansions for different values of μ_s .

1.2.2 Math Derivation

I want to calculate the initial stress, which is the initial shear stress at dip angle $\tau_{0,dip}$ and the initial normal stress at the dip angle $\sigma_{0,dip}$.

To detail this derivation, I begin with the definition of the prestress ratio:

$$R_{opt} = \frac{\tau_{0,opt} - \tau_{r,opt}}{\tau_{p,opt} - \tau_{r,opt}}$$

The prestress ratio R_{opt} is a function of the S -ratio (Equation 1.2.1) and is used to establish the initial stress level [20, 19].

By substituting the dynamic stress drop ($\Delta\tau_{opt} = \tau_{0,opt} - \tau_{r,opt}$) and the peak stress ($\tau_{p,opt} = c + \mu_s \cdot \sigma_{n^0,opt}$), Equation 1.2.2 becomes:

$$R_{opt} = \frac{\Delta\tau_{opt}}{c + \mu_s \cdot \sigma_{n^0,opt} - \tau_{0,opt} + \Delta\tau_{opt}}$$

Rearranging the terms yields:

$$(1 - R_{opt})\Delta\tau_{opt} = R_{opt} \cdot c + R_{opt} \cdot \mu_s \cdot \sigma_{n^0,opt} - R_{opt} \cdot \tau_{0,opt}$$

Following Appendix A4 from Aochi et al. [2], the shear and normal stresses ($\tau_{0,opt}$ and $\sigma_{n^0,opt}$) on a fault plane oriented at the optimal angle ϕ_{opt} are given by:

$$\begin{aligned}\tau_0 &= ds \cdot \sin(2\phi_{opt}) \\ \sigma_n^0 &= P - ds \cdot \cos(2\phi_{opt}) \\ \phi_{opt} &= \frac{\pi}{4} - 0.5 \cdot \arctan(\mu_s)\end{aligned}$$

(Note: Equation 1.2.2 holds true because for $\mu_s \in [0, 1]$, $2\phi_{opt} \leq 90^\circ$).

Using standard conventions [2], P and ds (the center and radius of the Mohr-Coulomb circle, respectively) are:

$$P = \frac{s_1 + s_3}{2}$$

$$ds = \frac{s_1 - s_3}{2}$$

Thus, $P = s_3 + ds$. Substituting these geometric relations back into Equation 1.2.2 gives:

$$(1 - R_{opt})\Delta\tau_{opt} = R_{opt} \cdot c + R_{opt} \cdot \mu_s \cdot s_3 + R_{opt} \cdot \mu_s \cdot ds - R_{opt} \cdot \mu_s \cdot ds \cdot \cos(2\phi_{opt}) - R_{opt} \cdot ds \cdot \sin(2\phi_{opt})$$

By replacing $\Delta\tau_{opt}$ using Equation 1.2.3 (derived below) and solving for ds , I obtain a quadratic equation in the form of:

$$a \cdot ds^2 + b \cdot ds + c = 0$$

where the coefficients are explicitly expanded as:

I define the angles as:

$$\alpha = \pi - 2\phi_{dip}$$

$$\beta = 2\phi_{opt}$$

$$a = a_1 + a_2$$

$$b = b_1 + b_2$$

$$c = c_1 + c_2$$

$$a_1 = (1 - R_{opt})(\sin \beta + \sin \beta \cdot \cos \alpha + \sin \alpha - \sin \alpha \cdot \cos \beta)$$

$$a_2 = -R_{opt}(\mu_s - \mu_s \cdot \cos \beta - \sin \beta + \mu_s \cdot \cos \alpha + \mu_s \cdot \cos \beta \cdot \cos \alpha + \sin \beta \cdot \cos \alpha)$$

$$b_1 = (1 - R_{opt})(\sin \beta \cdot s_3 + \Delta\tau - \Delta\tau \cdot \cos \beta - \sin \alpha \cdot s_3)$$

$$b_2 = -R_{opt}(2\mu_s \cdot s_3 + \mu_s \cdot s_3 \cdot \cos \alpha + c + c \cdot \cos \alpha - \mu_s \cdot \cos \beta \cdot s_3 - \sin \beta \cdot s_3)$$

$$c_1 = (1 - R_{opt})\Delta\tau \cdot s_3$$

$$c_2 = -R_{opt}(c \cdot s_3 + \mu_s \cdot s_3^2)$$

Solving this quadratic equation yields ds :

$$ds = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note that ds represents more than just the geometric radius of the Mohr-Coulomb circle; it carries significant physical meaning as the deviatoric stress (half the difference between the maximum and minimum principal stresses). As such, it must strictly be a positive value:

$$ds > 0$$

If $R = 1$, indicating the largest possible Mohr-Coulomb circle before rupture occurs, the relation simplifies to:

$$ds = \frac{-c - \mu_s \cdot s_3}{\mu_s - \mu_s \cdot \cos(2\phi_{opt}) - \sin(2\phi_{opt})}$$

1.2.3 Derivation of the Stress Drop ($\Delta\tau_{opt}$)

For a fault where the dip angle is $> 45^\circ$, I refer to the Mohr-Coulomb geometry outlined in Figure 1.9.

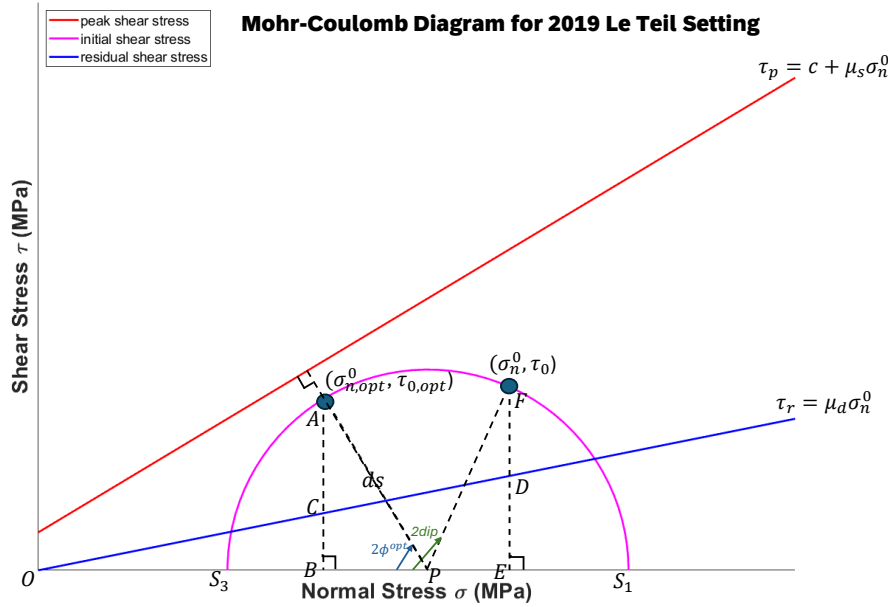


Fig. 1.9 Mohr-Coulomb geometric relationships used to derive stress drop for the 2019 Le Teil earthquake.

Using similar triangles $\triangle OCB$ and $\triangle ODE$, the ratios of their sides are equal:

$$\frac{DE}{CB} = \frac{OE}{OB}$$

This translates to the following stress differences:

$$\frac{FE - FD}{OE} = \frac{AB - AC}{OB}$$

Where the geometric segments map directly to my physical stress variables:

$$FE = ds \cdot \sin \alpha = ds \cdot \sin(\pi - 2\phi_{dip}) = ds \cdot \sin(2\phi_{dip})$$

$$FD = \Delta\tau_{dip}$$

$$OE = \sigma_{n^0,dip} = s_3 + ds + ds \cdot \cos \alpha = s_3 + ds - ds \cdot \cos(2\phi_{dip})$$

$$AB = ds \cdot \sin \beta = ds \cdot \sin(2\phi_{opt})$$

$$AC = \Delta\tau_{opt}$$

$$OB = \sigma_{n^0,opt} = s_3 + ds - ds \cdot \cos \beta = s_3 + ds - ds \cdot \cos(2\phi_{opt})$$

Substituting these into Equation 1.2.3 gives:

$$\frac{ds \cdot \sin(2\phi_{dip}) - \Delta\tau_{dip}}{\sigma_{n^0,dip}} = \frac{ds \cdot \sin(2\phi_{opt}) - \Delta\tau_{opt}}{\sigma_{n^0,opt}}$$

Isolating $\Delta\tau_{opt}$ ultimately provides the necessary expression for the stress drop:

$$\Delta\tau_{opt} = ds \cdot \sin(2\phi_{opt}) - \frac{(ds \cdot \sin(2\phi_{dip}) - \Delta\tau_{dip})(s_3 + ds - ds \cdot \cos(2\phi_{opt}))}{(s_3 + ds - ds \cdot \cos(2\phi_{dip}))}$$

1.2.4 Computing the Output Stress

Once ds is determined, the initial stress parameters at the optimal orientation angle ($\phi_{opt} < 45^\circ$) can be computed directly:

$$\tau_{0,opt} = ds \cdot \sin(2\phi_{opt})$$

$$\sigma_{n^0,opt} = s_3 + ds - ds \cdot \cos(2\phi_{opt})$$

$$\tau_{p,opt} = c + \mu_s \cdot \sigma_{n^0,opt}$$

$$\tau_{r,opt} = \tau_{0,opt} - \Delta\tau_{opt}$$

$$S_{opt} = \frac{1}{R_{opt}} - 1$$

Similarly, the initial stress parameters at the actual fault dip angle ($\phi_{dip} > 45^\circ$) are computed as:

$$\tau_{0,dip} = ds \cdot \sin(2\phi_{dip})$$

$$\sigma_{n^0,dip} = s_3 + ds + ds \cdot \cos(2\phi_{dip})$$

$$\tau_{p,dip} = c + \mu_s \cdot \sigma_{n^0,dip}$$

$$\tau_{r,dip} = \tau_{0,dip} - \Delta\tau_{dip}$$

$$S_{dip} = \frac{\tau_{p,dip} - \tau_{0,dip}}{\tau_{0,dip} - \tau_{r,dip}}$$

The remaining principal stresses can then be derived:

$$s_1 = s_3 + 2ds$$

$$s_2 = s_3 + 2 \cdot \mathbf{v} \cdot d\mathbf{s}$$

1.2.5 Estimating D_c

For a linear slip-weakening friction law, the critical slip distance (D_c) can be estimated using a standard scaling relation. Based on my kinematic slip model, I use D_{max} (the slip at each subfault converted from a regularized Yoffe-type source time function) and estimate D_c using the scaling relation proposed by Tinti et al. [18]:

$$D_c \propto \sqrt{\frac{T_{acc}}{\tau_R}} D_{max} \quad (1.3)$$

Assuming a linear scaling factor C_1 where $C_1 = 1$, Equation 1.3 becomes:

$$D_c = C_1 \sqrt{\frac{T_{acc}}{\tau_R}} D_{max} \quad (1.4)$$

where D_{max} is the final slip, T_{acc} is the positive acceleration time, and τ_R is the rupture duration. Figure 1.10 displays the interpolated D_c distribution across the fault plane.

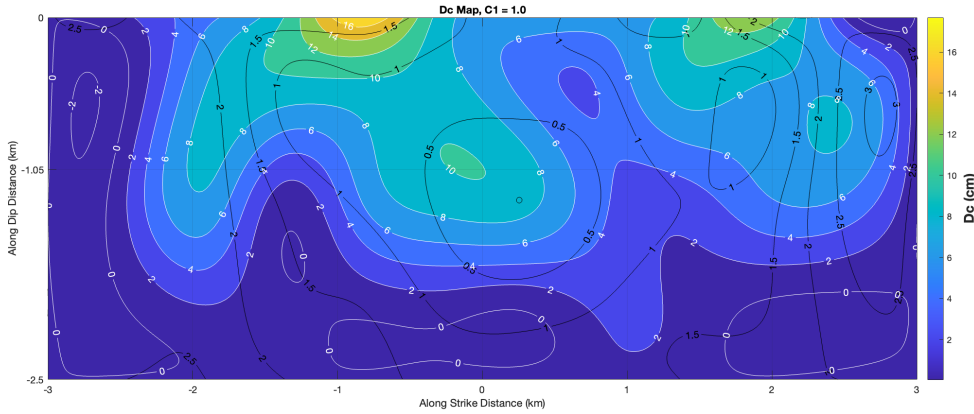


Fig. 1.10 Map of the interpolated D_c parameter across the fault plane (assuming $C_1 = 1$). The black contours denote rupture onset times, while the white contours represent the magnitude of D_c .

Alternatively, I can estimate D_c using other established methods. For instance, I could apply a simpler empirical scaling relation of $D_c = 0.4 \times D_{max}$ [16]. Another option is Mikumo's method [13], which defines D_c as the slip accumulated at the time of the peak slip rate. Because I already possess the final slip (D_{max}) and all the necessary parameters to construct the source time function, calculating D_c using these alternative approaches is straightforward.

1.3 Estimating Dynamic Parameters by Solving Elastodynamic Equations

Alternatively, I can estimate the dynamic parameters by solving the elastodynamic equations. Following a methodology similar to previous studies [6, 18], I obtain the stress evolution by imposing the kinematic slip-rate history as a fault boundary condition. By solving the forward elastodynamic equations, I determine the stress changes required to reproduce this imposed rupture evolution. This process allows me to estimate key dynamic parameters—specifically the critical slip distance (D_c), stress drop, and stress excess—directly from the stress-slip curves extracted at each subfault.

Figure 1.11 illustrates the theoretical traction-slip curve alongside the regularized Yoffe source time function and the corresponding traction evolution history. From the traction-slip curves, the three primary dynamic parameters are clearly defined: stress excess (referred to as strength excess in Tinti's original nomenclature), stress drop (dynamic stress drop), and critical slip distance (D_c).

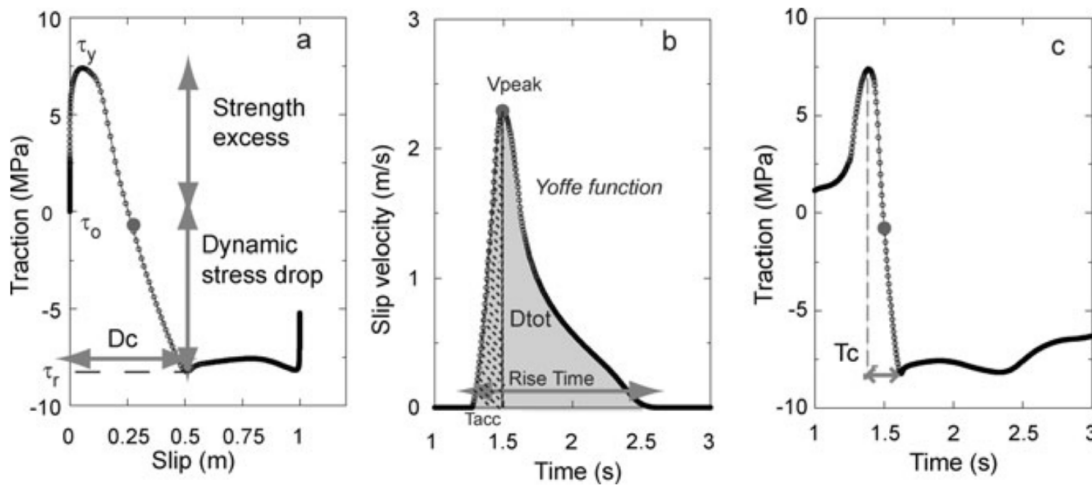


Fig. 1.11 Conceptual relationships between the regularized Yoffe slip-rate function, traction evolution, and the resulting traction-slip curve. source: Tinti et al. (2009) [16].

In Section ?? of the attached paper manuscript, I detail the mathematical derivation of how the initial stress and frictional parameters are estimated using the stress drop and stress excess as inputs.

Figure 1.12 illustrates the imposed slip-rate history of the kinematic model (left) alongside the resulting dynamic model solved by SeisSol (right). The middle-top panel compares the moment rate functions of these two models, while the middle-bottom panel compares their rupture onset times. The differences between the two models are distinct: the original kinematic model employs a triangular source time function and consists of only 60 subfaults (low spatial resolution). In contrast, the model solved by SeisSol utilizes a regularized Yoffe function and features over 50,000 subfaults, using a linear method for spatial interpolation.

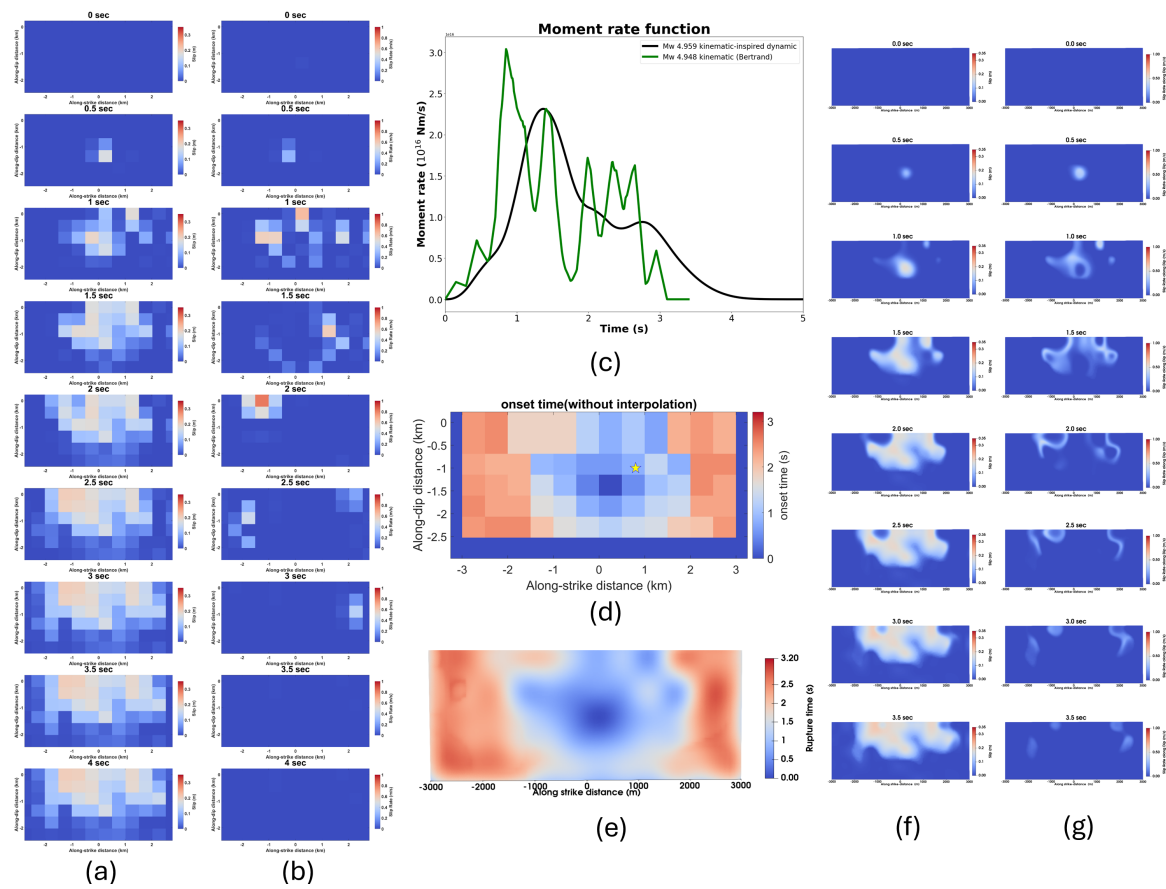


Fig. 1.12 Comparison of the kinematic model and the dynamic model solved by SeisSol, illustrating the slip-rate histories, moment rate functions, and rupture onset times.

1.4 Strain-Consistency Condition

1.4.1 Introduction

The results of dynamic rupture modeling strongly depend on a reliable estimation of the initial stress conditions. However, stress conditions at depth cannot be measured directly, and the prestress state before an earthquake is generally unknown. When detailed information is unavailable, a common approach is to assume a lithostatic stress condition [21], in which the vertical stress increases linearly with depth. Combined with Anderson’s theory of faulting, this assumption implies that, for a reverse-faulting regime, the vertical stress corresponds to the minimum principal stress, while the horizontal stress can be estimated using the Mohr–Coulomb criterion.

However, when this assumption of continuous vertical stress is combined with a layered velocity structure, it implicitly produces unphysical discontinuities in strain across layer boundaries [3]. This presents a physical inconsistency because strain should remain continuous across material interfaces, whereas stress may vary depending on local elastic properties.

To address this, Aochi and Tsuda (2023) [3] demonstrated a strain-consistent model for the 2019 Le Teil earthquake that properly accounts for a layered velocity structure. The strain-consistency constraint ensures that the strain in the upper layers does not exceed that of the lower layers. The constrained strain used in this formulation is deviatoric, calculated as the deviatoric stress $\left(\frac{1}{2}(\sigma_1 - \sigma_3)\right)$ divided by the shear modulus of each respective layer.

1.4.2 Layered Velocity Structure

Figure 1.13 and Table 1.1 illustrate and list the layered velocity model utilized in the dynamic modeling, adapted from [5]. Figure 1.13a displays a layered velocity model, mapped onto a 3D mesh alongside the fault slip from dynamic rupture.

Table 1.1 Velocity Model from Causse et al. (2021) [5]

Depth [km]	0.0	0.63	1.2	1.42	2.03	2.19	5.96
Density [kg/m ³]	2407	2688	2165	2465	2470	2667	2685
V_s [m/s]	2047	3645	1200	2291	2314	3457	3616

Figure 1.14 displays a clipped view of the computational mesh. This 3D volumetric mesh consists of unstructured tetrahedral elements of varying sizes that gradually coarsen with distance from the fault plane. I present the mesh as a clipped half-section to clearly reveal the internal detailing and spatial resolution of the elements. Throughout this study, I denote this central fault plane as the La Rouvière Fault (LRF).

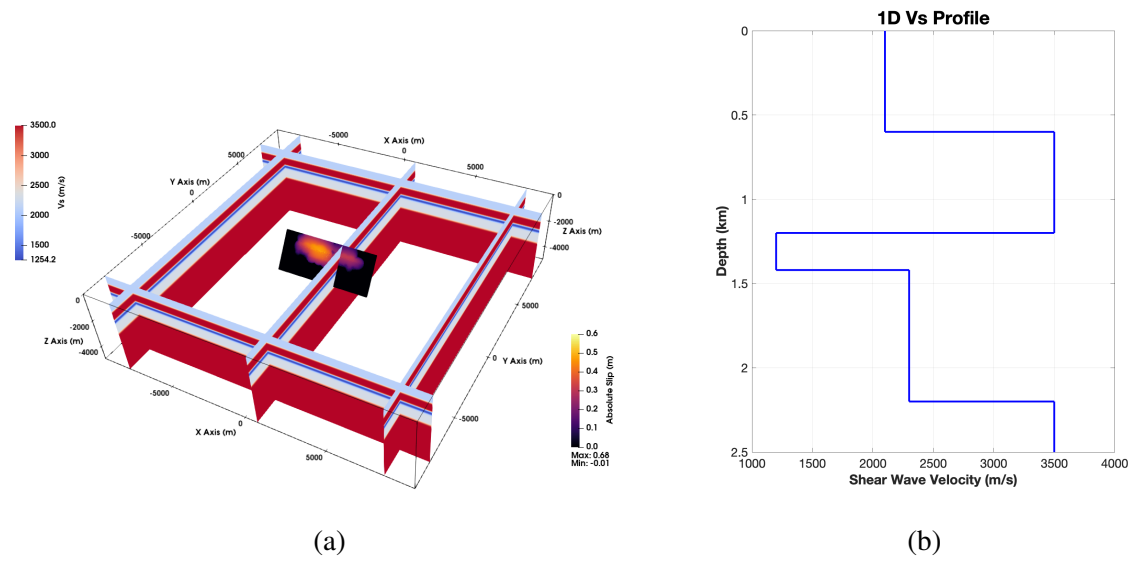


Fig. 1.13 (a) Display of the layered velocity model in a 3D mesh, plotted with the fault slip of a dynamic rupture model realization. (b) 1D layered velocity model from Causse et al. (2021) [5].

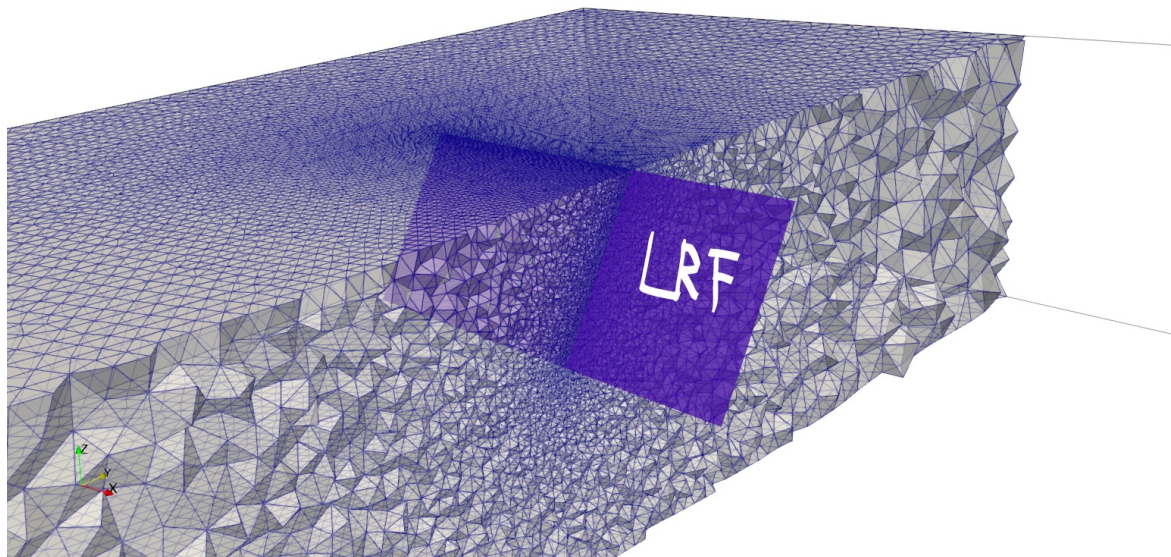


Fig. 1.14 Reconstruction of the La Rouvière Fault (LRF) within an unstructured tetrahedral computational mesh for the Le Teil area. The mesh is clipped to display the internal geometry and varying sizes of the elements.

1.4.3 Model Reproduction and Stress Profiles

I reproduced the model from Aochi and Tsuda (2023). The input parameters are summarized in Table 1.2, and the 1D layered velocity model is derived from Causse et al. (2021) [5].

Table 1.2 Input parameters for the model of Aochi and Tsuda (2023).

Model parameters							
dip = 50°	s_2 -ratio = 0.5	$R = 1$	$\alpha = 0$	$\mu_d = 0.33$	$\mu_s = 0.6$	cohesion = 5 MPa	
Depth [km]	0.00	0.63	1.20	1.42	2.03	2.19	5.96
ρ [kg/m ³]	2407	2688	2165	2465	2470	2667	2685
V_s [m/s]	2047	3645	1200	2291	2314	3457	3616

Figure 1.15 compares the maximum principal stress (S_1), minimum principal stress (S_3), and normal stress before and after applying the strain-consistency constraint. Before the strain-consistency check, the stress profiles are perfectly continuous with depth. However, after enforcing the strain-consistency condition, the principal and normal stresses become naturally discontinuous across the layer boundaries, reflecting the changes in the local material properties.

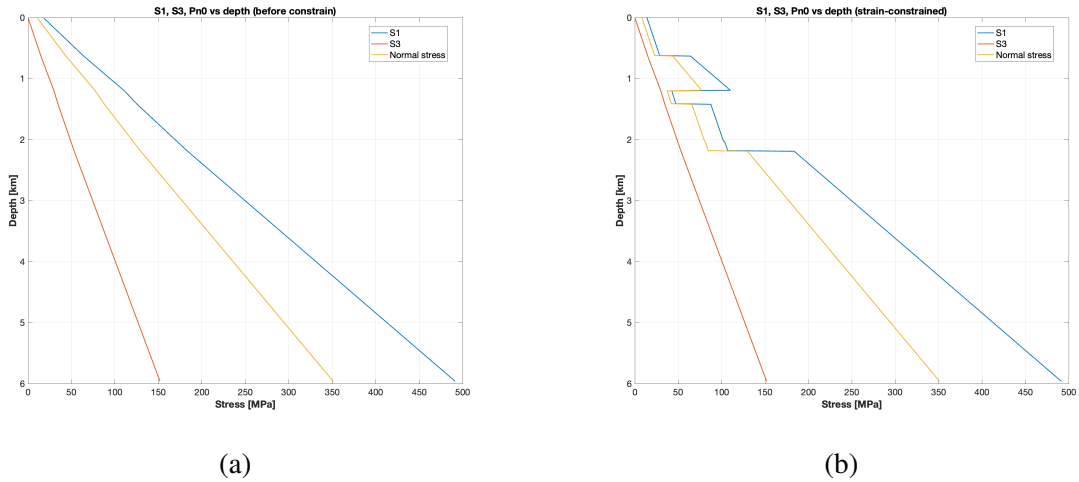


Fig. 1.15 S_1 , S_3 , and normal stress as a function of depth. (a) Before applying the strain-constrained condition. (b) After applying the strain-constrained condition.

Similarly, Figure 1.16 compares the peak (τ_p), initial (τ_0), and residual (τ_r) shear stresses before and after applying the constraint. Just like the principal stresses in Figure 1.15, these shear stress profiles are continuous prior to the check but become discontinuous once strain consistency is enforced. Notably, at certain layer boundaries, the applied constraint causes the initial stress to drop below the residual stress. This creates a zone of negative stress drop.

Figure 1.17 demonstrates the impact of these stress changes on the S -parameter distribution. After applying the strain-consistency constraint, the S -parameter profile exhibits sharp discontinuities and becomes negative at certain depths. A negative or very large S -parameter

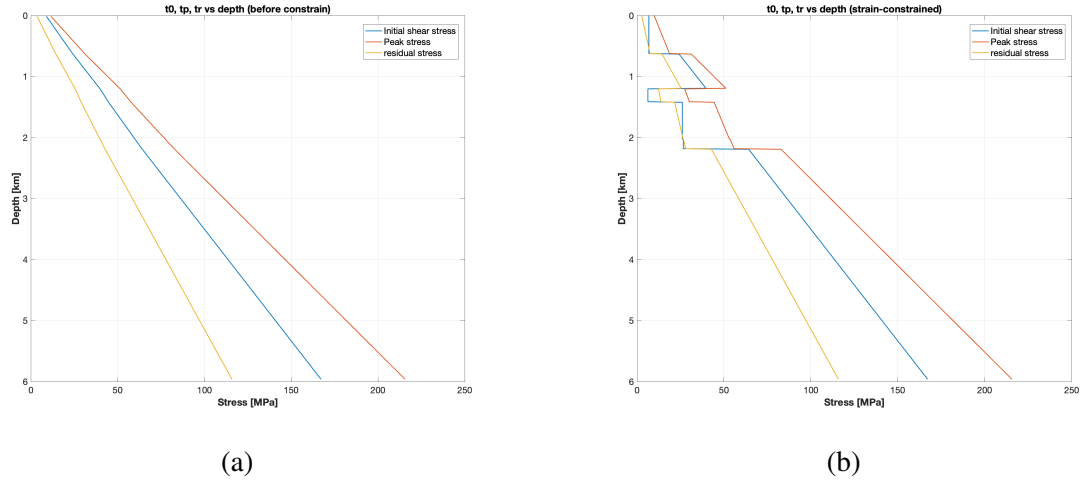


Fig. 1.16 τ_0 , τ_p , and τ_r as a function of depth. (a) Before applying the strain-constrained condition. (b) After applying the strain-constrained condition.

indicates the presence of a strong stress barrier. This means that the necessary correction of the loading stresses effectively creates deeper zones that would hinder or completely stop dynamic rupture propagation.

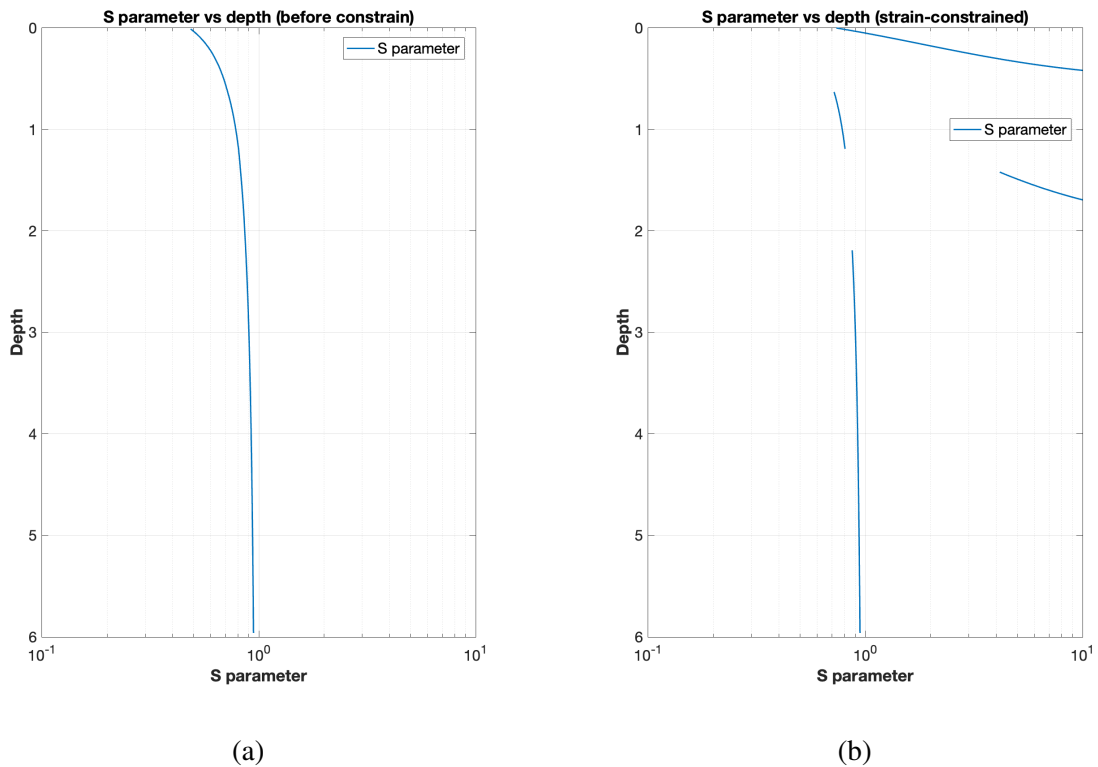


Fig. 1.17 S-parameter distribution before and after applying the strain-constrained condition.

Finally, Figure 1.18 compares the strain profiles before and after the constraint alongside the 1D layered rheological model (shear modulus). By ensuring that the strain is physically

consistent across layers, the initially continuous vertical stress profiles are forced to become discontinuous at depth, meaning the stress at depth is loaded unevenly.

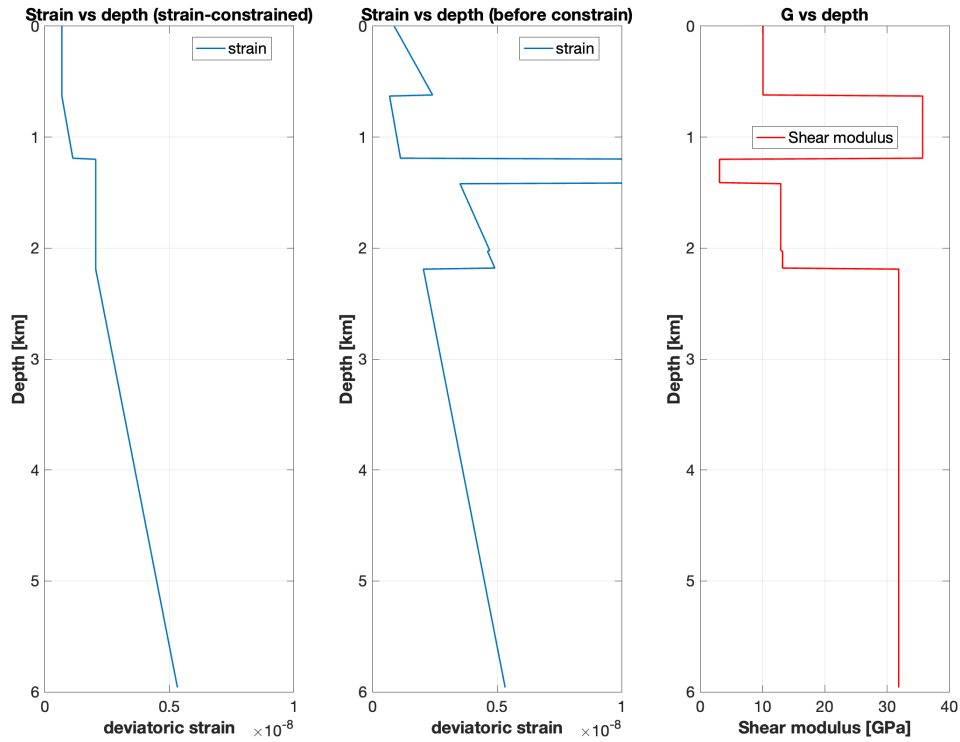


Fig. 1.18 From left to right: strain vs. depth (after applying the strain-constrained condition), strain vs. depth (before applying the strain-constrained condition), and shear modulus vs. depth.

1.5 Linear Slip-Weakening Friction

The linear slip-weakening friction law is a widely used simplification of physical frictional behavior during an earthquake rupture [11, 14, 1]. To fully define this model, four key parameters are required: cohesion, the static friction coefficient (μ_s), the dynamic friction coefficient (μ_d), and the critical slip-weakening distance (D_c).

Figure 1.19 illustrates the connection between these linear slip-weakening friction parameters and the fault's stress state [15]. In this model framework, μ_s represents the peak static friction before failure, μ_d represents the residual dynamic friction during sliding, D_c is the critical slip distance over which the friction drops from static to dynamic, and cohesion defines the inherent shear strength of the fault at zero normal stress.

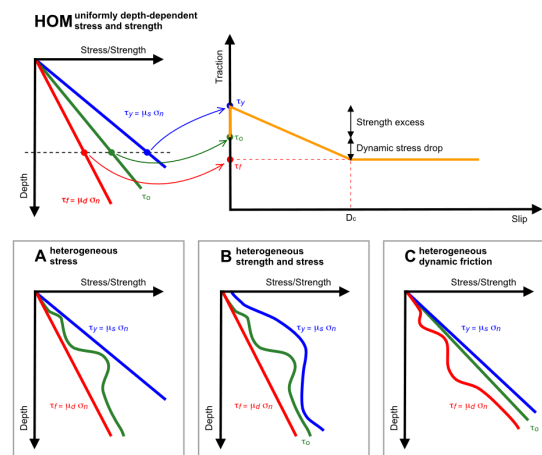


Fig. 1.19 The top panel illustrates the relationship between the linear slip-weakening friction model and the corresponding stress parameters. The bottom panel outlines three possible approaches to introducing heterogeneity into the stress models: (1) heterogeneous initial stress, where the peak and residual stresses remain depth-dependent; (2) heterogeneous peak and initial stresses, resulting in a heterogeneous stress excess; and (3) heterogeneous residual stress, resulting in a heterogeneous stress drop. Note that in the original nomenclature used by Tinti et al., peak stress is referred to as yield stress, and residual stress is referred to as final stress. Figure source: [15].

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